Good-Citizen Lottery*

Duk Gyoo Kim†

July 7, 2026

Abstract

This paper examines the Good-Citizen Lottery, a budget-balanced institutional mechanism designed to collectively minimize public bads by using penalties from violators to fund a reward for one randomly-selected prosocial agent. Contrary to standard voluntary contribution models where cooperation decays in large groups, our game-theoretic model predicts the asymptotic stability of good-citizen behavior. This is driven by the budget-neutral nature of the lottery: as the group size grows, the decreased probability of winning the lottery is offset by the endogenously increasing prize pool (funded by more violators). Experimental evidence is consistent with this theoretical prediction: the proportion of good behavior is robust and weakly increases with group size. Finally, we show that this institutional success is further enhanced by a behavioral channel: individuals who subjectively overestimate small winning probabilities find the uncertain reward relatively more appealing, sustaining compliance levels above the risk-neutral benchmark.

JEL Classification: D73, C91, H41, K42

Keywords: Lottery, Public bads, Imperfect monitoring, Laboratory experiments

*This work was supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea (NRF-2023S1A5A8080738). I thank Syngjoo Choi, Binglin Gong, Luca Henkel, Biung-Ghi Ju, Heechun Kim, Jeong-Yoo Kim, Hyoeun Park, and Christoph Schlom for their comments and seminar participants at Sungkyunkwan University and Yonsei University and conference participants of the 2024 East Asia Game Theory Conference, the 2024 Korea Experimental and Behavioral Economics Association Workshop, 2025 Korea Allied Economic Associations Annual Meeting, 2025 Asia-Pacific Meeting of the Economic Science Association, 2025 World Congress of the Econometric Society, 2026 Asia Meeting of the Econometric Society, 2026 Meeting of the Society for Social Choice and Welfare for their comments.

†School of Economics, Yonsei University, South Korea. Email: kim.dukgyoo@yonsei.ac.kr

1 Introduction

Good citizenship is often, if not always, unpaid. Consider a driver who wants to stop at a building that has no available parking spaces. Although it may cause various forms of social costs, parking illegally for two minutes might avoid detection, so he may be tempted to park in front of the building and sneak away from the scene shortly. Alternatively, the driver could take the time to find a legal parking spot, walk the extra distance, and pay the parking fee, all without receiving explicit acknowledgment for doing the "right thing." Good citizenship encompasses a wide range of behaviors, but this paper focuses on actions that are costly to the individual yet do not harm society.[1] In this framework, good citizens are those who voluntarily abstain from contributing to the production of public bads, such as reducing greenhouse gas emissions at home, recycling responsibly, refraining from littering, and paying for public transportation when monitoring is minimal.

The literature on public bads minimization has mostly focused on penalizing bad[2] citizens, while little attention has been paid to ways to encourage good citizenry. Another apparent solution to minimize social costs is to increase government spending on monitoring capacities or offsetting social costs, but this raises subsequent questions about the cost-effectiveness of government intervention. Having these issues in mind, my broader research question is how to facilitate good-citizen behaviors without imposing additional government expenses. This paper proposes a novel method to collectively minimize the production of public bads without requiring an external budget: a good-citizen lottery, hereinafter a *citizen lottery*. Under the status quo, each citizen decides how to behave based on their own benefits and costs. Under the lottery, one randomly-selected good citizen receives a lottery payment whose prize is funded by imperfectly-collected penalties from bad citizens.[3] Would this mechanism effectively decrease the proportion of bad citizens? If so, how does the mech-

[1] A concept of good citizens may be innocuously extended to law-abiding business owners and taxpayers. For example, the recent legalization of cannabis in some U.S. states forces legal cannabis sellers to compete against the opportunistic benefits of being an illegal seller. News reports indicate that California's first cannabis company filed for bankruptcy largely because they could not compete with illegal weed stores, which constitute more than 70 percent of the market in Los Angeles.

[2] Here the 'bad' citizen does not necessarily mean that the citizen's behavior is morally bad. Throughout this paper, I refer to behavior that renders a positive expected payoff to the individual but imposes a social cost via negative externality. Without proper monitoring capacity and penalization, this 'bad' behavior can be understood as economically 'rational' behavior.

[3] For three days in September 2010, the Swedish Road Safety Organization and Volkswagen trialed a 'speed camera lottery,' whereby drivers who drove under the speed limit were entered to win a lottery prize funded by fines paid by speeding drivers (AAP FactCheck. "Sweden's speed camera lottery hit a red light years ago." December 12, 2019. Available at aap.com.au. Last accessed: June 29, 2026.) To the best of my knowledge, this trial is the only known implementation of the budget-balanced good-citizen lottery described in this paper.

anism perform as the population size grows?

To understand the effect of the citizen lottery, I consider a game where citizens simultaneously choose one of two actions. One action, corresponding to a rent-seeking (or bad-citizen) behavior, involves a large benefit with some probability and a penalty with the complementary probability. The other action, corresponding to a good-citizen behavior, comes with no benefits, but one of the good citizens is randomly selected to get an award whose value is the sum of the penalties collected by a fraction of bad citizens.

With a reasonable set of parameters, the equilibrium probability of good actions is asymptotically stable as the population size grows. The intuition relies on the budget-neutral nature of the mechanism. As the population grows, two forces interact. On one hand, a larger number of good citizens implies that the probability of winning the lottery approaches zero (a dilution effect). On the other hand, a larger population with the same fraction of good citizens implies a larger absolute number of bad citizens, which scales up the collected penalties and, consequently, the prize pool (a valuation effect). These two forces—the decreasing probability of winning and the increasing value of the prize—offset each other. This result stands in contrast to the typical prediction for voluntary contributions to public goods, where individual incentives to contribute asymptotically shrink to zero as the number of beneficiaries grows (Andreoni, 2006). In the citizen lottery, because the reward scales endogenously with the number of violators, the incentive to maintain good citizenship remains robust even in large populations. Furthermore, an increase in monitoring capacity increases the equilibrium probability of good actions, implying that this mechanism complements, rather than substitutes, standard enforcement efforts.

Experimental findings are consistent with these theoretical predictions. As the group size increases, the proportion of (neutrally framed) good-citizen behaviors remains stable and does not decay. Moreover, the *level* of good citizenship observed in the experiment is consistently higher than the risk-neutral theoretical benchmark. This "level shift" is primarily driven by subjects with a stronger tendency toward probability weighting: individuals who subjectively overestimate small winning probabilities find the lottery reward relatively more appealing than the standard model predicts. These insights reinforce the viability of the citizen lottery: the institutional structure ensures stability against group size, while psychological biases (probability weighting) enhance the overall compliance level. A supplementary online experiment with substantially larger group sizes (from 50 to 200) provides corroborating evidence, confirming that the proportion of good-citizen behaviors remains robust and well above theoretical predictions even as the group size scales up.

The remainder of this paper is organized as follows. The following subsection provides an overview of the relevant literature. Section 2 describes the model. Section 3 presents the theoretical analyses, and Section 4 describes the laboratory experiment. Section 5 reports the experimental findings, and Section 6 discusses various issues, including the supplementary evidence from an online experiment with larger group sizes. Section 7 concludes.

1.1 Literature Review

The use of lotteries in nonstandard contexts has gained considerable attention in recent years as an innovative mechanism to address diverse societal challenges. Kim (2021) explores the use of vaccination lotteries to examine how they promote vaccination uptake compared to a lump-sum subsidy, while Kim (2023) introduces a penalty lottery framework that endogenously makes citizens reveal their willingness to produce public bads. Gerardi et al. (2016) and Duffy and Matros (2014) examine the relationship between penalties for not turning out to vote and turnout lotteries, demonstrating their potential to incentivize voter participation. Kearney et al. (2010) and Filiz-Ozbay et al. (2015) advocate for savings lotteries as a tool to encourage financial discipline among individuals who did not have savings accounts. Morgan (2000) and Morgan and Sefton (2000) propose and experimentally examine the efficacy of lotteries as a method to fund public goods, showing that such mechanisms can overcome free-rider problems. The "regret lottery" that pays a randomly-selected employee who did not park in the facility worked as an effective tool to reduce car use (Gneezy, 2023), while inducing regret in others who were selected but disqualified due to car use. Other innovative applications include Björkman Nyqvist et al. (2018), who examine the effectiveness of a lottery to promote safer sexual behavior, and Volpp et al. (2008) and Levitt et al. (2016), who explore lotteries as a means to foster desirable habits, such as improving health outcomes and student achievement. Collectively, these studies underline the versatility of lotteries in motivating behaviors across a range of settings.

To the extent that the minimization of public bads corresponds to the provision of public goods, this study is related to the extensive experimental literature on public goods provision (Isaac et al., 1984; Isaac and Walker, 1988; Andreoni, 1990; Charness and Yang, 2014). However, a key distinction of the good-citizen lottery is its asymptotic behavior. A standard voluntary contribution mechanism (VCM) model without warm glow utility predicts that the voluntary contributions asymptotically decrease as the population size grows (Andreoni, 2006). In contrast, the budget-neutral design of the citizen lottery creates a countervailing force, the scaling prize, that stabilizes cooperation in large groups. Additionally, unlike standard VCM studies that often rely on framing effects to sustain cooperation (Andreoni, 1995), this study adopts an abstract and neutral framing to isolate the strategic incentives of the mechanism itself.

The most closely related work is Fabbri et al. (2019), who conducted a field experiment on a bus-ride lottery aimed at improving compliance with purchasing public transportation tickets. While their study offers valuable insights, there are three significant differences. First, this paper ensures budget balancedness, which imposes a natural constraint on the expected benefits of good-citizen behaviors and creates the endogenous stability described above. Second, rather than comparing the lottery mechanism to a no-lottery baseline, this study investigates how the group size and monitoring capacity influence the effectiveness of the lottery. Finally, it delves into individual heterogeneities that drive good-citizen behaviors, utilizing a behavioral channel (probability weighting)[4] to explain why compliance levels exceed risk-neutral predictions. These distinctions position the current study as a novel contribution to the literature on institutional design and the incentivization of prosocial behaviors.

2 Model

This section considers the simplest possible model. Although there are many avenues to extend this model, the fundamental tradeoff between two actions of each decision maker would still remain unchanged.

Suppose there are n citizens, indexed by $i \in \{1, \ldots, n\} \equiv N$, making a decision simultaneously.[5] Each citizen chooses one of two actions: one action is to safely abide by law, denoted by S, and another action is to violate it to accrue private benefits, V. Thus, the action space is $A = \{S, V\}^n$.

Citizen i accrues a benefit of acting V, $B_i > 0$.[6] Although the citizen of acting V can be monitored and fined F, the monitoring capacity is limited. Specifically, with probability $p \in (0,1)$, action V is monitored so that the citizen's payoff is $B_i - F < 0$, while with probability

[4] A recent policy experiment relying on this bias is the "Lucky Yatra" pilot introduced by Indian Railways in 2025. The scheme converted standard train tickets into lottery entries to incentivize fare compliance. While the initiative was designed to exploit the overestimation of small probabilities, subsequent reports indicated mixed operational results due to low public awareness. See *Storyboard18* (June 21, 2025), "Cannes Lions Grand Prix winner 'Lucky Yatra' fails to deliver for Indian Railways."

[5] This assumption of simultaneity does not mean that all the citizens must act at the same time. Rather, this assumption parsimoniously captures the imperfect information about other citizens' actions.

[6] Alternatively, one can consider a cost of acting S, which essentially works the same.

$1-p$, she enjoys the full benefit of B_i. The payoff of choosing S is 0 for the 'most' cases, but when selected as a winner of the lottery, it is kF, where k is the number of players who chose V and were caught and fined for violating the law.[7]

I made two parametric assumptions here. I assume that the monitoring capacity is determined by the external agency or the government, so p is exogenously given. Also, I assume $B_i - pF > 0$ for all i, so bad behavior is beneficial in expectation. The model becomes trivial otherwise: If $B_i - pF \leq 0$ for some i, such citizens will find that V is strictly dominated by S.

While citizens may naturally derive varying levels of benefit $B_i > 0$ from the violation, for the remainder of the theoretical analysis we impose the assumption of homogeneous benefits, such that $B_i = B$ for all $i \in \{1, \ldots, n\}$. This transition to a homogeneous benchmark yields tractable analytical solutions for the symmetric mixed-strategy Nash equilibrium, allowing us to cleanly isolate the core mechanics of the citizen lottery.[8]

Before we analyze the model, I illustrate the key tradeoff with an example of three homogeneous citizens. It can be easily understood as a variation of a coordination game. One clear prediction is that (S,S,S) can never be an equilibrium: Given that two other citizens play S, the expected payoff of playing V is $B - pF$, while the expected payoff of playing S is $0 (= 0 + 0F)$ since no penalties are collected. Since $B - pF > 0$, the citizen has an incentive to deviate to V. (V,V,V) cannot be an equilibrium either under some parametric conditions. Given that two other citizens play V, the expected payoff of playing S is $2pF$, where $2p$ is the expected number of citizens who got a penalty. So, as long as $2pF > B - pF$, or $\frac{B}{pF} < 3$, the citizen has an incentive to deviate to play S. (In other words, if $\frac{B}{pF} > 3$, the unique Nash equilibrium is (V,V,V).) When $\frac{B}{pF} < 3$, a symmetric mixed-strategy Nash

[7] An alternative institutional formulation could distribute the collected fines equally among all compliant citizens. Under risk neutrality, a proportional distribution and a winner-take-all lottery yield the identical expected payoff. However, empirical and experimental evidence suggests that the winner-take-all lottery format is behaviorally more effective at inducing participation. For example, in the context of public health compliance, Kim (2021) demonstrates that vaccine lotteries generate a significantly stronger behavioral response than mathematically equivalent, deterministic lump-sum transfers. Therefore, we focus on the lottery structure as it represents the more potent, policy-relevant mechanism.

[8] While the assumption of homogeneous B simplifies the formal exposition, the qualitative comparative statics remain robust to heterogeneous benefits. In this framework, a citizen's expected payoff from playing S depends primarily on the *aggregate* probability of compliance among the rest of the population (which determines the expected prize pool and the expected number of winners). Because the incentive structure shifts universally, changes to the group size n or monitoring capacity p will drive the aggregate compliance rate in the same direction, even if individual threshold strategies vary based on specific B_i.

equilibrium is for each citizen to play S with probability δ, where

$$\delta = \frac{3-\sqrt{\frac{4B}{pF}-3}}{2}.$$

Note that we assume $B - pF > 0$, so $\frac{4B}{pF} - 3 > 1$ and $\delta < 1$. Note also that $\delta > 0$ as long as $\frac{B}{pF} < 3$. Because we already established the condition $\frac{B}{pF} < 3$ to prevent (V, V, V) from being the unique pure-strategy Nash equilibrium, this condition is naturally satisfied in this intermediate parameter range. In words, if the monitoring capacity p is sufficiently high to deter universal violation ($p > \frac{B}{3F}$), but not high enough to completely deter the action V outright ($p < \frac{B}{F}$), there is a symmetric equilibrium where citizens play a good-citizen behavior with some positive probability.

We focus on the symmetric mixed-strategy Nash equilibrium for two primary reasons. First, in a decentralized, anonymous setting without communication, citizens lack the coordination devices necessary to sustain an asymmetric pure-strategy equilibrium (where exactly k citizens play S and $n-k$ play V). The symmetric mixed strategy thus serves as the most natural behavioral benchmark. Second, this formulation seamlessly captures *aggregate* behavior. If we relax the homogeneity assumption, this mixed probability δ can be interpreted as the aggregate proportion of compliance emerging from a population where individuals play pure strategies based on heterogeneous private types.

It is worth noting that if the population could coordinate on an asymmetric pure-strategy equilibrium, the qualitative comparative statics would remain identical. Instead of the continuous probability δ smoothly increasing with group size, the aggregate number of citizens playing S would exhibit a corresponding discrete, step-wise increase as n grows. Thus, the core theoretical prediction that will be shown in the next Section—good-citizen behavior does not decay at scale—is robust to the choice of equilibrium concept.

3 Theoretical Analysis

For notational simplicity, suppose there are $n+1$ citizens (instead of n citizens) so that each citizen's decision takes into account n other citizens' decisions. A profile $(\delta)_{i=1}^{n+1}$ is a symmetric mixed-strategy Nash equilibrium if the expected payoff of playing S:

$$\sum_{i=1}^{n}\binom{n}{i}\delta^{n-i}(1-\delta)^i\frac{ipF}{n+1-i}$$
$$=\binom{n}{0}\delta^{n}(1-\delta)^0 0F+\binom{n}{1}\delta^{n-1}(1-\delta)\frac{pF}{n}+\binom{n}{2}\delta^{n-2}(1-\delta)^2\frac{2pF}{n-1}+\cdots+\binom{n}{n}\delta^{0}(1-\delta)^n npF \quad (1)$$

is equal to the expected payoff of playing V:

$$B-pF. \quad (2)$$

The compact summation in Equation 1 carries a straightforward intuitive structure. The first term, $\binom{n}{i}\delta^{n-i}(1-\delta)^i$, is the binomial probability that exactly i out of the other n citizens choose to play V. The second term, $\frac{ipF}{n+1-i}$, represents the payoff to the compliant individual in that specific event: the expected prize pool generated by i violators (ipF) is awarded to the compliant citizen with a winning probability of $\frac{1}{n+1-i}$ (since the focal citizen competes with $n-i$ other citizens playing S).

Therefore, $\delta\in[0,1]$ such that

$$\sum_{i=1}^{n}\binom{n}{i}\delta^{n-i}(1-\delta)^i\frac{ipF}{n+1-i}=B-pF$$

is each citizen's equilibrium probability of playing S. Let $b:=\frac{B}{pF}-1\in(0,n)$ denote the normalized excess benefit relative to the expected cost. Note that $b<n$ is required to ensure an interior equilibrium where the probability of playing S is strictly positive. Then, we have the following simpler form:

$$b=\sum_{i=1}^{n}\binom{n}{i}\delta^{n-i}(1-\delta)^i\frac{i}{n+1-i}:=R(\delta,n) \quad (3)$$

Equation 3 demonstrates the tradeoff of playing S versus V: Playing V renders the normalized excess benefit, but as more people play V, the expected size of the good-citizen lottery prize increases, leaving the opportunity cost of not playing S larger. This equation works as a basis for the comparative statics. Note that the left-hand side of Equation 3 is constant in n, and the right-hand side of it is constant in p, the comparative statics on these parameters are straightforward. Before analyzing the equilibrium strategy and examining comparative statics, it is useful to observe that $R(\delta,n)$ has nice properties.

Lemma 1. *$R(\delta,n)=\sum_{k=1}^{n}(1-\delta)^k$. $R(0,n)=n$, $R(1,n)=0$, and $R(\delta,n)$ is monotone decreasing in $\delta\in[0,1]$.*

Proof: See Appendix.

Thanks to Lemma 1, the equilibrium probability of good-citizen behavior δ is such that

$$b=\sum_{k=1}^{n}(1-\delta)^k, \tag{4}$$

which is analytically tractable. The first comparative static regards the changes in n.

Proposition 1. *Let $b\in(0,n)$ be a constant and n be an integer. Let $\delta^*(n)\in(0,1)$ be the solution to $b=\sum_{k=1}^{n}(1-\delta)^k$. $\delta^*(n)$ is unique, and it is strictly increasing in n.*

Proof: See Appendix.

Proposition 1 demonstrates the most desirable feature of the citizen lottery: As the number of citizens gets larger, the probability of choosing S, or the proportion of good citizens, increases.

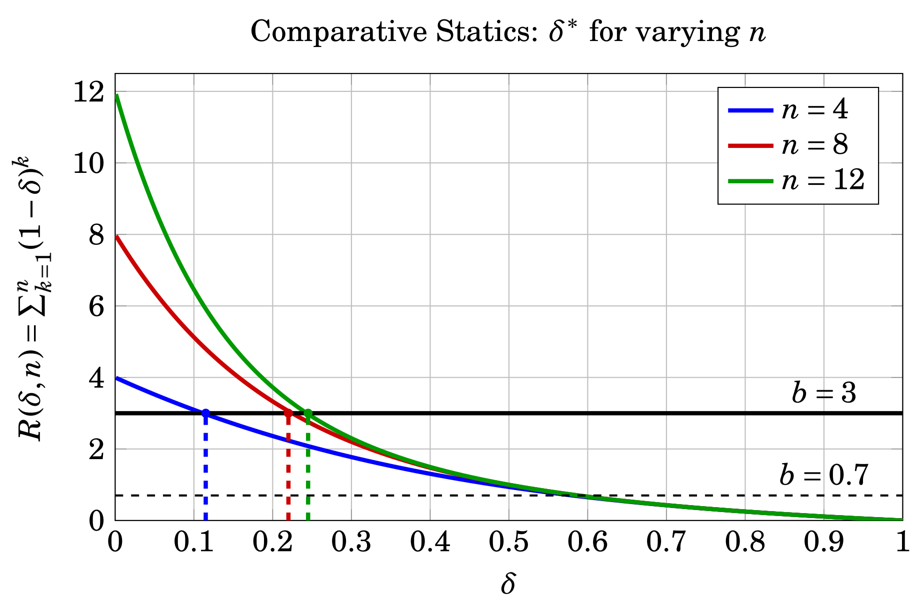


Figure 1: A larger population increases the proportion of good citizens.

Notes: This figure plots the equilibrium probability of playing the good-citizen action S (δ^*) with respect to the group size (n), illustrating the monotonic comparative static derived in Proposition 1. The theoretical prediction is plotted using the fixed parameter $b=\frac{B}{pF}-1=3$.

Figure 1 illustrates Proposition 1 and demonstrates additional properties. The x-axis of the figure represents the value of δ ranging from 0 to 1, while the y-axis represents the value of the left- and right-hand sides of Equation 4. The left-hand side of it is invariant to the changes in n and δ, so it is depicted as a horizontal line: I arbitrarily chose $b = 3$ for illustration. The right-hand side of it, $R(\delta, n)$ decreases in δ from n to 0 (Lemma 1). The crossing point between the decreasing curve and a horizontal line describes the equilibrium strategy δ^*. As illustrated in Figure 1, $\delta^*(n)$ increases when n increases from 4 to 12. The intuition relies on understanding the two opposing forces that dictate the expected benefit of playing S. On one hand, given the same δ, a larger population dilutes the chance of winning the lottery. On the other hand, it increases the expected prize pool, as there are more potential violators to fund it. The reason the upward force dominates is driven by a finite-population asymmetry. From the perspective of a citizen choosing S, they are an entrant in the lottery but are explicitly not part of the set of other citizens generating fines. Consequently, when the population grows by one additional citizen, it adds both possible prize funding and possible dilution. These two effects do not exactly cancel out; the expected growth in the prize pool slightly outweighs the dilution, yielding a strictly positive residual (represented algebraically by the $(1-\delta)^{n+1}$ term). It is important to note that Proposition 1 establishes the monotonic increase of the equilibrium probability δ. Since a probability of each citizen playing S leads to the expected frequency, not ex post frequency, this dictates the strategic pressure rather than guaranteeing a deterministic increase in the empirical fraction of good citizens in every small sample.

Another noticeable observation is that when b is between 0 and 1, the equilibrium δ^*, would be almost stable: See the three downward-sloping curves almost coincide with a horizontal dashed line depicting $b = 0.7$. A formal argument for this observation is stated in Proposition 2.

Proposition 2. *Let $b \in (0,1)$ be a constant. The sequence $\{\delta^*(n)\}_{n=1}^{\infty}$, defined by the solution to $b = \sum_{k=1}^{n}(1-\delta)^k$, is a Cauchy sequence that converges to $\delta^*(\infty) = \frac{1}{1+b}$. Furthermore, for any $n \geq 1$, the incremental change is bounded by $\delta^*(n+1) - \delta^*(n) < b^{n+1}$, implying that for $b \in (0,1)$, the sequence stabilizes exponentially fast.*

Proof: See Appendix.

Propositions 1 and 2 provide desirable properties of the citizen lottery. First, when the normalized excess benefit of V is high, a larger population size leads to a larger proportion of good citizens. This implies that the citizen lottery effectively deters many citizens from playing V even though doing so is highly beneficial. Second, when the benefit of V is still positive but not that high, the proportion of good citizens is stable as the population size grows. This stability result may help a policymaker to make a reasonable prediction without concerning too much about population uncertainty. I will use this property when choosing parameters for the laboratory experiment.

Two clarifications about the scope of Proposition 2 are in order. First, the limiting value $\lim_{n\to\infty}\delta^*(n) = \frac{1}{1+b}$ obtains for *any* $b > 0$: It follows from equating b with the convergent geometric series $\sum_{k=1}^{\infty}(1-\delta)^k = \frac{1-\delta}{\delta}$, which requires only $\delta \in (0,1)$. The restriction $b \in (0,1)$ is used only to bound the *rate* of convergence through $\delta^*(n+1) - \delta^*(n) < b^{n+1}$. Consequently, for $b \geq 1$ equilibrium compliance still rises monotonically in n (Proposition 1) toward the same limit $\frac{1}{1+b}$, but it approaches that limit gradually rather than stabilizing almost immediately.

The second comparative static regards the changes in p.

Proposition 3. $\frac{d\delta^*}{dp} > 0$ *for* $b \in (0,n)$.

Proof: The right-hand side of Equation 4 is constant in p and decreasing in δ. Since the left-hand side, $b = \frac{B}{pF} - 1$, decreases in p, δ^* that equates both hands increases. □

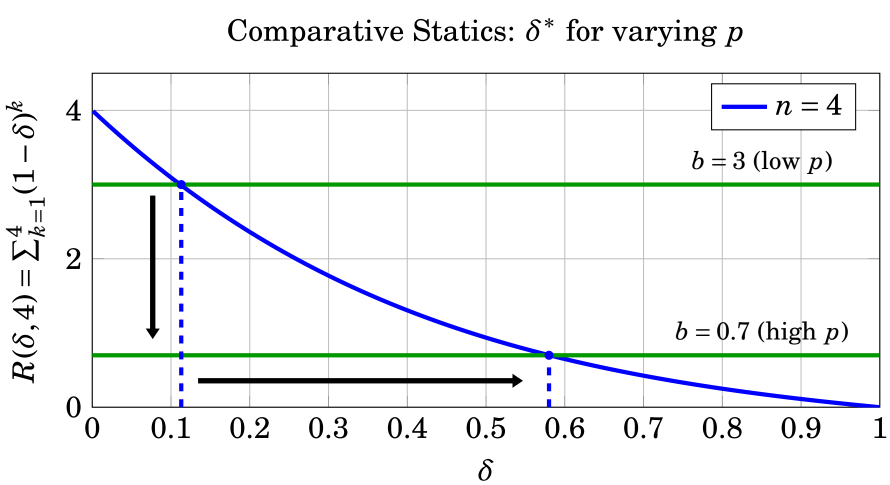


Figure 2: Monitoring capacity encourages good-citizen behaviors.

Notes: This figure plots the equilibrium probability of playing the good-citizen action S (δ^*) as a function of monitoring capacity (p), illustrating the comparative static derived in Proposition 3. The theoretical prediction is plotted using fixed parameters n, B, and F, so that the changes in p govern the changes in $b \in \{0.7, 3\}$.

Figure 2 illustrates Proposition 3. As b decreases to b' due to the increase in the monitoring capacity, the equilibrium probability of playing S increases. Since the left-hand side of Equation 4 does not involve n, this holds for all n. Two dashed lines show how the equilibrium probability of playing S increases for $n = 4$ when b is changed from 3 to 0.7. Although it may seem too straightforward that the better monitoring capacity decreases the excessive benefit of playing V, it is worth noting that the increase in $\delta^*(p)$ happens only when $b \in (0,n)$. When the normalized excess benefit of V is greater than n, that is, when p is too low, B is too large, or F is too small, the equilibrium probability of playing S converges asymptotically to zero.

Risk preferences and subjective probability weighting are worth mentioning as two important additional factors that could affect the citizens' decisions. Regarding risk preferences, it is not immediately straightforward which action is "riskier" for intermediate group sizes. On the one hand, playing V is a persistent binary gamble, yielding a payoff of B with probability $1-p$ and $B-F$ with probability p. On the other hand, playing S exposes the citizen to the uncertainty of the lottery, where a payoff of 0 is highly likely but a positive payoff of kF (where k, the number of monitored violators, is itself a random variable) is possible.[9] Consequently, for the intermediate group sizes considered in our experiment, the theoretical relationship between risk aversion and the likelihood of playing S is ambiguous and depends heavily on structural assumptions. However, as the group size n gets larger, the probability of winning the good-citizen lottery approaches zero, even as the potential prize pool grows. Combined with the standard diminishing marginal utility of a risk-averse player, the utility gained from this massive potential prize increases at a relatively slower rate than the probability of winning decreases. Consequently, the expected utility of playing S approaches that of a certain payoff of zero. In contrast, the payoff of V remains a volatile binary gamble regardless of group size. Thus, a highly risk-averse agent would eventually prefer the asymptotic certainty of S over the persistent variance of V as $n \to \infty$. Because the net effect is theoretically ambiguous for our specific experimental group sizes, I do not formulate a directional prediction regarding risk preferences, but instead treat it as an exploratory empirical question.

Another factor the risk-neutral model does not explicitly incorporate is the heterogeneous tendency to subjectively overestimate small probabilities (Tversky and Kahneman, 1992). Individuals often purchase lotteries because they weight the chance of winning more

[9] To illustrate, consider an extreme case where there are only two citizens. Given the other citizen's probability of playing S is δ, action S is associated with a random payment of 0 with probability $1-(1-\delta)p$ or F with probability $(1-\delta)p$. Meanwhile, action V is associated with a random payment of B with probability $1-p$, and $B-F$ with probability p. Which one gives a larger expected payoff for a risk-averse citizen depends on the size of B, F, the monitoring capacity p, and the risk-aversion parameter.

heavily than the objective probability dictates (Blau et al., 2020). Such tendencies manifest in the 'Favorite–Longshot Bias' in betting markets (Thaler and Ziemba, 1988; Snowberg and Wolfers, 2010), where bettors accept a negative expected value for the chance of a high payoff. Extending this logic to the citizen lottery, subjects who exhibit subjective probability weighting should be more attracted to the small chance of winning the accumulated fines.

Claim 1. *The greater a subject's tendency to overestimate small probabilities, the higher their likelihood of playing the good-citizen action S.*

4 Experiment

To examine how the citizen lottery works in practice, I conducted a laboratory experiment, varying two key parameters, n and p. I consider the theoretical predictions and claim summarized in Section 3 as hypotheses for the experiment and design the experiment to test them clearly.

4.1 Experimental Design

An experiment is designed as follows: In each session of $N \in \{18, 20\}$ participants, they play 12 similar games, wherein each subject is randomly assigned to a group whose size is $n \in \{3, 6, 9, 18\}$ when $N = 18$ or $\{2, 5, 10, 20\}$ when $N = 20$.[10] To be explicit, in every round, all N participants in the session are perfectly partitioned into mutually exclusive groups of size n. For example, when $N = 18$ and $n = 6$, the 18 participants are partitioned into three distinct groups of 6; no participants sit out. Their task is to choose one of the two items: a white ball and a box.[11] Unwrapping the box, a subject gets a red ball with probability $p \in \{0.3, 0.5\}$ and a blue ball with probability $1 - p$. By getting a blue ball, a subject earns a payoff of $B + m$. A red ball is associated with a payoff of $B - F + m$, where $m > 0$ is the base payoff to guarantee the participant's minimum earnings to be positive. Choosing the white ball earns m. On top of that, one of the group members who chose the white ball is randomly selected to get an additional payoff of kF, where k is the number of the members who got the red ball within the group of n.

[10] It would be ideal to have all the sessions with the same number of participants, but due to unexpected no-shows I had to prepare for two different contingencies.

[11] To avoid any unobservable responses to the framed narratives, I consider abstract framing. A white ball and a box, respectively, correspond to actions S and V in section 2, but in any part of the experiment instructions, neither normative nor judgmental descriptions were used. Considering that subjects might have some baseline utilities on good citizenry, my finding could be the lower bound of the estimate.

Round	1	2	3	4	5	6	7	8	9	10	11	12
n (when $N=18$)	6	9	18	3	9	6	18	3	9	6	3	18
n (when $N=20$)	5	10	20	2	10	5	20	2	10	5	2	20
Choosing a box $\Rightarrow$ Red ball with prob p; Blue ball with prob $1-p$. Choosing a white ball $\Rightarrow$ Randomly selected one additionally earns kF.												
$p=0.3$ or $p=0.5$												

Table 1: Experimental Design, Varying Session Sizes $N \in \{18, 20\}$

Table 1 summarizes the experimental design. Each round comes with a different group size in the mixed order as shown in Table 1. The participants were told that they would know how many subjects form a group at the beginning of each round, without knowing the group sizes of the following rounds. To control for the potential order effect, the same mixed order is used for all sessions. For example, in the first round of every session with 18 participants, they were told that the group size is 6. Between subjects, there are two treatments in terms of the monitoring capacity p. I call the treatment with $p = 0.3$ as P03, and P05 is denoted accordingly. After completing the 12 rounds, risk preferences and subjective probability weighting tendencies are elicited via a simple survey.[12] Post-experimental survey includes some individual characteristics, which would be used as control variables later. Finally, one of the 12 rounds is randomly selected to be paid.

I set the parameters to be tightly aligned with the theoretical predictions in Section 3, and to guarantee the theoretical minimum payment to be reasonably close to a typical show-up payment. The experiment currency unit used in this experiment is tokens, and I set the exchange rate at 1 token to 100KRW (about 0.07USD), the base payoff m to be 100 tokens, and B and F to be 140 and 200 tokens, respectively. This means, when a subject chooses a box, it comes with $240(=140+100)$ tokens with probability $1-p$, or with 40(=

[12]To measure subjects' risk preferences, they were asked to answer two questions requiring a choice between a certain payoff and a binary gamble with a higher expected value. The second question was dynamically adjusted based on their answer to the first. A subject is coded as high risk-taking ('HighR') if they hypothetically chose the risky gamble (a 50% chance of 30,000 KRW, 0 otherwise) over the certain payoff (10,000 KRW) in the first question, and 'LowR' otherwise. The second question refined the risk threshold conditional on the first response; because it was dynamically adjusted rather than an absolute measure, I use the first (unconditional) choice for a clean dichotomous split. To measure the tendency to subjectively overestimate small probabilities, subjects completed a non-incentivized survey following the main experiment. They were asked to state their willingness to pay (WTP) to purchase a lottery with a 5%, 15%, and 30% chance of winning, as well as their WTP to avoid a 5%, 15%, and 30% chance of losing. A subject is coded as exhibiting the probability weighting tendency ('PrW') if they demonstrated a disproportionate valuation of small probabilities—specifically, if $\mathrm{WTP}_{5\%}/\mathrm{WTP}_{15\%} > 5/15$ or $\mathrm{WTP}_{15\%}/\mathrm{WTP}_{30\%} > 15/30$.

$140 - 200 + 100$) tokens with probability p. Choosing a white ball comes with a payoff of 100 tokens, and in case of the lottery winner, $200 * k$ tokens are added, where k is the number of the red balls appeared in the group. Given those parameters, the normalized excess benefit is $b = \frac{B}{pF} - 1 = \frac{4}{3}$ in P03 ($p = 0.3$) and $b = \frac{2}{5}$ in P05 ($p = 0.5$). Substituting these into the asymptotic limit $\lim_{n\to\infty} \delta^*(n) = \frac{1}{1+b}$ derived in Section 3, the equilibrium proportion of good-citizen (white-ball) choices converges to $\frac{1}{1+4/3} = \frac{3}{7} \approx 0.43$ in P03 and to $\frac{1}{1+2/5} = \frac{5}{7} \approx 0.71$ in P05. The two treatments differ, however, in how this limit is approached over the experimental group sizes. Because $b = \frac{2}{5} < 1$ in P05, Proposition 2 implies that $\delta^*(n)$ stabilizes exponentially fast, so the predicted compliance rate is essentially flat at $\frac{5}{7}$. In P03, by contrast, $b = \frac{4}{3} > 1$ lies outside the fast-stabilization regime, but Proposition 1 still guarantees that $\delta^*(n)$ increases monotonically in n, rising toward $\frac{3}{7}$ across the group sizes we consider. This increasing-in-P03 versus stable-in-P05 pattern is exactly the treatment-specific prediction that the experiment is designed to test.

4.2 Hypotheses

Corresponding to the propositions and claim summarized in Section 3, I primarily investigate the following three testable hypotheses. For notational consistency, I call the action of choosing a white ball in the experiment as playing S.

Hypothesis 1. *In P03, the fraction of subjects playing S is increasing with the group size. In P05, the fraction of subjects playing S is relatively stable with the group size.*

Hypothesis 2. *The fraction of subjects playing S is greater in P05 than in P03.*

Hypothesis 3. *The subjects with a stronger tendency to overestimate small probabilities would prefer to play S.*

While the theoretical model assumes risk neutrality, it is natural to ask how risk aversion might impact the decision to play S. Formally deriving equilibrium behavior with risk aversion would require imposing many structural assumptions regarding the distribution of heterogeneous risk preferences and players' beliefs about others' risk attitudes. However, the basic intuition is that the effect of risk aversion is ambiguous for intermediate group sizes, because both actions involve uncertainty. Playing V exposes the subject to a binary gamble (yielding B with probability $1 - p$ and $B - F$ with probability p), while playing S exposes the subject to the uncertainty of the citizen lottery. As the group size n grows sufficiently large, the probability of winning the citizen lottery approaches zero. Provided with standard diminishing marginal utility, the payoff of S converges to a nearly certain payoff of 0. The payoff of V, however, remains a volatile gamble. Thus, for a highly risk-averse player, the persistent variance of V becomes unattractive relative to the asymptotic certainty of S, suggesting that risk aversion might eventually encourage good-citizen behavior in extremely large groups. For the intermediate group sizes considered in our experiment, however, the theoretical net effect is ambiguous, or at least, vulnerable to many structural assumptions. Therefore, rather than formulating a directional hypothesis, I include standard risk-elicitation tasks in the experiment to treat the role of risk preferences as an exploratory research question and to ensure our main results are robust to heterogeneous risk attitudes.

A few remarks on the hypotheses are worth mentioning. These hypotheses are not meant to be normative: I do not claim that the experimental findings must be consistent with what theory says. Instead, those must be considered as theoretical benchmarks. We can learn more from what is different from theory, not from what is as predicted. For example, the relationship between risk preferences and likelihood of playing S is theoretically ambiguous, so I set the hypothesis of no relationship between them to learn what the experimental findings guide us. It is also worth mentioning the role of Hypothesis 2. Because a higher p mechanically lowers the normalized excessive benefit of playing V, $\frac{B-pF}{pF}$, this comparative static is straightforward. Therefore, Hypothesis 2 serves primarily as a validation check to confirm that subjects comprehend the incentive structure and the experimental findings are not driven by other factors such as experimenter demand effect.

Notably, the experimental design omits a 'no-lottery' baseline treatment, which may be understood as a standard fine-only system. Because the model assumes $B - pF > 0$, the risk-neutral theoretical prediction for compliance without a lottery is precisely zero. Given that baseline behavioral noise (e.g., altruism, confusion) in such standard environments is already well-documented, this experiment prioritizes testing the novel group-size dynamics and scalability unique to the budget-balanced institution.

4.3 Experimental Procedure

All sessions were conducted via the real-time online mode using Zoom. Upon arrival at the designated Zoom meeting, subjects' sign-up information were checked, and then they were instructed to rename their display name with two random alphabet letters. Profile images were disabled so that the Zoom meeting environment does not make any one of the participants distinctive. Each received a web link to the standalone experimental pages created by LIONESS (Live Interactive ONline Experimental Server Software). To induce public knowledge on the information contained in the instructions, the same instructions were presented on the participants' display, and the experimenter read them aloud. After all questions were addressed in Zoom, the participants answered comprehension check quizzes, and they started the session only when every participant passed the quizzes. The experiment was conducted at Sungkyunkwan University in South Korea and the original instructions were conveyed in Korean.[13] Table 2 summarizes the distribution of sessions, participants, and decision-level observations across the two monitoring treatments. With a total of 8 sessions (4 sessions of each of P03 and P05), the 150 participants generated 1,800 decision-level observations (12 rounds per participant).

Treatment	Sessions with $N = 18$	Sessions with $N = 20$	Total Participants	Total Observations
P03 ($p = 0.3$)	3	1	74	888
P05 ($p = 0.5$)	2	2	76	912
Total	5	3	150	1,800

Table 2: Summary of Experimental Observations

To minimize any dynamic effects from the previous decision rounds, the participants were told that at the beginning of every round, the entire subjects in the session were randomly shuffled and regrouped. The subjects earned on average 18,800KRW (about 14USD) with the minimum earnings of 5,000KRW and the maximum of 75,000KRW. Due to administrative restrictions regarding the cash payments to the subjects, a Starbucks e-gift card whose balance corresponds to the subject's earning was sent mobile.

5 Results

This section presents the experimental findings regarding the hypotheses developed in the previous section. The first set of findings concerns changes in the proportion of good-citizen behavior (playing S, or choosing a white ball) in response to changes in group size and monitoring capacity.

Figure 3 presents the main results. The x-axis represents the group size. For illustrative simplicity, group sizes are labeled categorically: L (size 3 in session 18; size 2 in session 20),

[13] The instructions translated in English are available in the Appendix B. The original Korean version is available upon request.

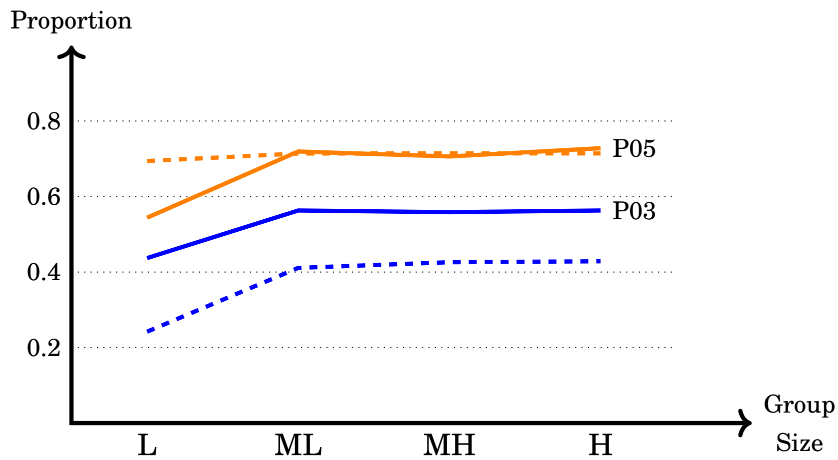


Figure 3: Proportion of playing S

Notes: Solid lines represent the observed experimental data, while dashed lines represent the risk-neutral theoretical predictions. Blue lines indicate the low-monitoring treatment (P03, $p = 0.3$) and orange lines indicate the high-monitoring treatment (P05, $p = 0.5$). The x-axis labels represent categorical group sizes: L (Low: size 2 or 3), ML (Medium-Low: size 5 or 6), MH (Medium-High: size 9 or 10), and H (High: size 18 or 20). Confidence intervals and standard errors for these observations are reported in Table 3.

followed by ML, MH, and H.[14] The y-axis represents the proportion of subjects who played action S (chose a white ball). Solid lines represent the observed experimental data, while dashed lines of the corresponding color represent the theoretical benchmark, derived by numerically solving the symmetric mixed-strategy equilibrium condition for each discrete group size n given the treatment parameters with $N = 18$. In P03, the theory predicts that the proportion of playing S increases with group size (from 24.2% to 42.86%). The experimental data shows a significantly higher average level of cooperation (near 50%) than the theoretical benchmark ($p = 0.003$).[15] Visually, the data exhibits a "parallel shift" upwards from the theoretical prediction. In P05, the theory predicts that the proportion of playing S should remain stable around 71.43%. The observed average level (67.43%) is not significantly different from this benchmark ($p = 0.109$). However, contrary to the theoretical prediction of stability, the proportion of S seems to exhibit an increase with group size.

To evaluate this trend formally, I conduct a qualitative test of the directional prediction

[14]This labeling is for illustrative purposes only. All regression results reported utilize group size as a continuous variable.

[15]This level comparison tests whether the observed proportion of S differs from the risk-neutral theoretical prediction, using a regression of the individual choice minus the theoretical benchmark on a constant with standard errors clustered at the individual level. It is distinct from the slope test reported in Table 3, which evaluates the trend in group size.

by running a linear regression of S choices on group size, with standard errors clustered at the individual level. Table 3 reports full estimation details. In the low-monitoring treatment (P03), the empirical proportion of S visually exhibits a weak upward trend that tracks the theoretical benchmark (Figure 3). However, the standard errors are slightly larger than those in the high-monitoring treatment (P05), rendering the empirical slope statistically indistinguishable from zero ($p = 0.148$). Therefore, while the qualitative pattern does not reject the model, the data lack sufficient statistical power to definitively confirm the predicted sensitivity to group size in the low-monitoring condition. In the high-monitoring treatment (P05), the proportion of S exhibits a statistically significant increase with group size.

Table 3: The Effect of Group Size and Treatment on the Likelihood of Playing S

	Dependent Variable: Choice of S (1 = Yes, 0 = No)					
	(1) P03 Only	(2) P03 Only	(3) P05 Only	(4) P05 Only	(5) Pooled	(6) Pooled
Group Size (n)	0.0064	0.0057	0.0080**	0.0087**	0.0072**	0.0074**
	(0.0044)	(0.0049)	(0.0036)	(0.0038)	(0.0028)	(0.0030)
Treatment (P05 = 1)					0.1435***	0.1189***
					(0.0370)	(0.0391)
Probability Weighting		0.1546**		0.0119		0.1048*
		(0.0740)		(0.1003)		(0.0589)
Risk-Taking Tendency		-0.0489		-0.1569***		-0.1037**
		(0.0606)		(0.0524)		(0.0404)
Constant	0.4724***	0.4224**	0.6016***	0.8542***	0.4648***	0.5537***
	(0.0438)	(0.1837)	(0.0397)	(0.2592)	(0.0348)	(0.1439)
Demographic Controls	No	Yes	No	Yes	No	Yes
Observations	888	756	912	840	1,800	1,596
R^2	0.0058	0.0295	0.0114	0.0390	0.0299	0.0507

Notes: Standard errors clustered at the individual level are reported in parentheses. Columns (1) and (2) restrict the sample to the low-monitoring treatment (P03). Columns (3) and (4) restrict the sample to the high-monitoring treatment (P05). Columns (5) and (6) pool the data across treatments. Demographic controls in Columns (2), (4), and (6) include gender and age; observations are dropped in these specifications due to subjects who preferred not to report their age and gender.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Overall, the experimental evidence provides mixed support for the sharp, treatment-specific comparative statics predicted by the model and stated in Hypothesis 1. However, the data strongly confirm the broader qualitative prediction of non-decay: contrary to standard public goods games, the proportion of subjects choosing the good-citizen action does not collapse as group size scales up.

Result 1. *In P05, the fraction of subjects playing S aligns with the theoretical benchmark in level, but unexpectedly increases with group size. In P03, the fraction of subjects playing S is significantly higher than the theoretical prediction, and the predicted increasing trend is not statistically significant.*

Hypothesis 2 concerns the response to changes in monitoring capacity. As expected from the mechanical structure of the model, the proportion of good-citizen behavior in P05 is significantly larger than in P03 ($p < 0.001$), supporting Hypothesis 2. This result acts as a necessary sanity check, confirming that subjects reacted appropriately to the core economic incentives before evaluating the more complex group-size and behavioral dynamics.

Result 2. *The proportion of subjects playing S is significantly greater in P05 (high monitoring) than in P03 (low monitoring).*

Given the between-subject design, Result 2 cannot be attributed to experimenter demand effects. This leaves an open question regarding P03: What explains the 'level shift,' where subjects play S significantly more often than the risk-neutral theory predicts? I first test Hypothesis 3 to see if subjective probability weighting can account for this behavioral premium. As reported in Table 3, it does.

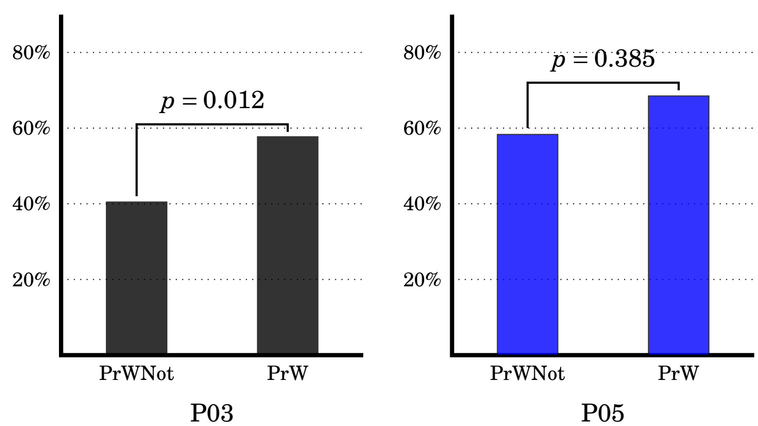


Figure 4: Proportion of Ball Choices By Probability Weighting Tendency

Notes: 'PrW' indicates subjects who exhibited a tendency to subjectively overestimate small probabilities in the post-experiment elicitation, while 'PrWNot' indicates subjects who did not exhibit this tendency.

Result 3. *In P03, subjects with a stronger tendency to overestimate small probabilities play S significantly more often, consistent with Hypothesis 3.*

Figure 4 breaks down the proportion of S by the subjects' tendency to overestimate small probabilities. I categorized subjects into two groups based on their self-reported willingness to pay for fictitious lotteries (gains) and insurance (losses), adapting questions from Rieger et al. (2017). A higher willingness to pay for low-probability events indicates a stronger tendency toward probability weighting.

Result 3 is particularly important for the scalability of the citizen lottery. P03 represents a "low enforcement" environment where the rational incentive to be a good citizen is weak. While the game-theoretic model correctly predicts that compliance will not decay with group size, it under-predicts the absolute level of compliance. This result provides evidence that the "Favorite–Longshot" bias acts as a behavioral multiplier, driving citizens to behave prosocially at rates higher than the risk-neutral benchmark predicts.

While probability weighting successfully explains a substantial portion of the observed level shift, it is natural to ask whether standard risk aversion might also play a role. As discussed in Section 4, I treat the effect of risk preferences as an exploratory question, since the theoretical prediction for intermediate group sizes is ambiguous. I therefore test whether heterogeneous risk preferences can serve as an alternative explanation for the level shift.

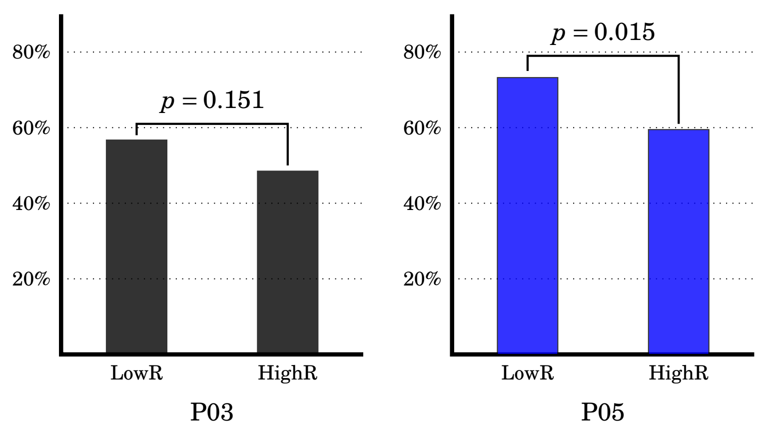


Figure 5: Proportion of Ball Choices By Risk Preference

Notes: 'HighR' indicates subjects who exhibited a tendency to take risks in the post-experiment elicitation, while 'LowR' indicates subjects who did not exhibit this tendency.

Figure 5 examines playing S in relation to risk preferences. Subjects with a higher risk-taking tendency play S less in both treatments, though the difference is not statistically significant in P03. These findings indicate that for the group sizes and parameters considered in this experiment, heterogeneous risk preferences do not drive behavior. Crucially, this exploratory finding confirms that our main treatment effects (Results 1 and 2) are robust and not confounded by underlying variations in risk aversion.

Result 4. *Subjects with higher risk aversion tend to play S more frequently, though the difference is statistically insignificant in P03.*

Because risk aversion and subjective probability weighting can manifest similarly in observed behavior, Table 3 includes both measures simultaneously. By controlling for baseline risk preferences in the multivariate regression, we isolate the significant effect of probability weighting on compliance and the insignificant effect of risk preference in P03.

Result 4 provides insight into the nature of the uncertainty subjects face. Action S involves strategic uncertainty (the prize depends on others' actions), while action V involves exogenous risk (the penalty depends on monitoring probability p). If risk aversion were primarily driven by an aversion to strategic uncertainty, risk-averse subjects would avoid S. The fact that they prefer S suggests that the aversion to the direct penalty risk in V dominates the concern for strategic uncertainty in S.[16]

6 Discussions

In this section, I address some issues not discussed in the main body of the paper.

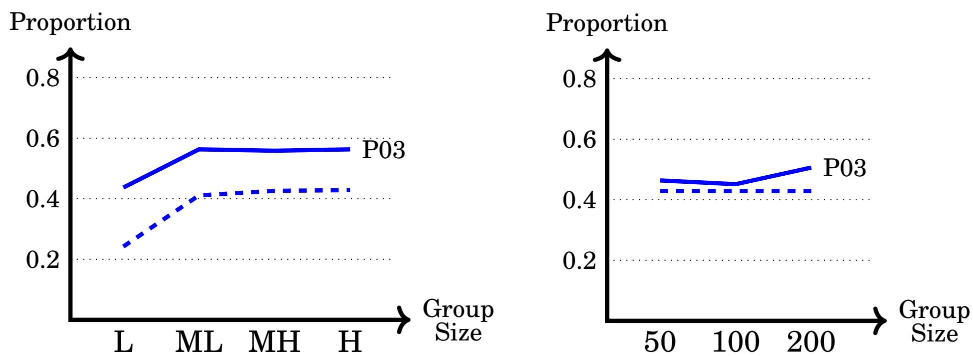
6.1 Supplementary Evidence from an Online Experiment

A common concern regarding laboratory experiments is that the group size, intended to represent a community or local society, is often too small. In the laboratory experiment, the largest group size was 20. Although this group size is sufficiently large for the equilibrium probability of choosing S to approximate the equilibrium in the infinite population limit, some may question whether the non-decreasing pattern observed in the lab is merely an

[16] Regarding strategic uncertainty, if the monitoring probability p were unknown to citizens, ambiguity-averse individuals might strictly prefer S to avoid the ambiguous downside risk associated with V. While this speculation is beyond the scope of the current paper, the interaction between ambiguity aversion and citizen lotteries remains a valuable avenue for further investigation.

artifact of the small scale. To address this concern and test the asymptotic stability predicted by the theoretical model, I conducted an additional online experiment on Prolific in February 2025 using substantially larger groups.[17]

The online experiment followed a similar procedure to the P03 treatment of the laboratory study, with three key differences: (1) decisions were made asynchronously, (2) participants were recruited from a representative sample of the United States, and (3) most importantly, the group sizes were 50, 100, and 200. After consenting to participate, subjects read the same instructions as in the laboratory experiment—choosing either a box or a white ball in each period, with varying payoffs associated with each decision. Participants were informed that their actual payment would be determined only after all group members had completed the study. At the start of each period, participants were randomly assigned to one of the group sizes: 50, 100, or 200. The order of group sizes was randomized. Following their decision-making, participants completed post-experiment survey questions. A total of 401 subjects participated.



(A) Laboratory

(B) Online Extension

Figure 6: Proportion of playing S.

Notes: Panel (A) depicts the laboratory experiment ($n \leq 20$). The x-axis labels denote categorical group sizes (L=Low, ML=Medium-Low, MH=Medium-High, H=High). Panel (B) depicts the asynchronous online extension ($n \in \{50, 100, 200\}$) with exact group sizes labeled. Because the populations and experimental modes differ, the absolute levels of compliance are not directly comparable across panels; rather, the online extension serves as an independent robustness check for large group sizes.

[17] This experiment was pre-registered on AsPredicted.org (registration #208968) under the same title, 'Good-Citizen Lottery'. Since the online experiment was conducted after the laboratory data had been collected, I explicitly stated that it serves as an extension of the laboratory findings. I also note for transparency that the pre-registration document reflects an earlier stage of the theoretical development, hypothesizing standard VCM decay. The formal budget-balanced model presented in Section 2, which predicts stability, was fully developed subsequent to this pre-registration.

Figure 6 presents the results. Panel A reproduces the laboratory data (P03) for reference, while Panel B displays the online extension. Because the online extension utilizes an entirely different subject pool and experimental mode, the absolute baseline level of compliance cannot be directly compared to the laboratory data. Furthermore, because group sizes in the online experiment are exceptionally large relative to the total sample of 401 subjects, there are too few independent group-level observations (e.g., only two independent groups of $n = 200$) to conduct adequately powered formal regression analyses. Therefore, the online results are presented as descriptive, qualitative evidence. Crucially, however, this within-sample comparison confirms the theoretical prediction of stability: there is no evidence of a collapse in the proportion of participants choosing the good-citizen action as group size drastically scales up.

6.2 Modeling Assumptions

This subsection addresses two core modeling assumptions made throughout the paper: the normative desirability of the bad-citizen behavior, and the assumption of citizen homogeneity which necessitates our focus on a mixed-strategy equilibrium.

I assumed that $B_i - pF > 0$ for every citizen i. This assumption was made to simplify the analysis and to ensure that the theoretical predictions without the citizen lottery are straightforward (i.e., universal non-compliance). It is, of course, possible to assume that citizens are distributed over an interval $[\underline{B}, \overline{B}]$, with a value $B^* \in [\underline{B}, \overline{B}]$ such that citizens with $B_i \leq B^*$ are innately good (finding $B_i - pF \leq 0$). Introducing such innately good citizens would lower the incentives for strategic individuals to act as good citizens, as the presence of 'innate' winners adds pure dilution to the winning probability without contributing potential fine revenue to the prize pool. As long as the proportion of these innate citizens is relatively small, strategic citizens ($B_i > B^*$) still face a prize pool that scales with the number of violators, preserving the mechanism's general robustness. However, if the variance in B_i heavily skews toward these low values, this dilution could eventually dominate the delicate positive residual identified in Proposition 1, defining the theoretical boundary where the monotonic stability might break down.

The assumption of citizen homogeneity is another important consideration. In assuming that there are no innately good citizens, I set $B_i = B$ for all i. However, other factors—particularly wealth and household income—may influence decision-making. It is well documented that lower-income households often spend a larger proportion of their income on lottery tickets. While the citizen lottery is distinct from a typical lottery (as there is no purchase fee), if the behavioral response to the "lottery" framing mimics that of commercial lotteries, the mechanism might exert a stronger deterrent effect on lower-income households. Although this potential unintended consequence—shifting the burden of good citizenship onto poorer individuals—is purely theoretical and has not been substantiated in this study, it remains an important consideration for welfare analysis.

A structural limitation of the homogeneous benchmark model is its reliance on a symmetric mixed-strategy Nash equilibrium, which requires citizens to be perfectly indifferent between compliance and violation. As is standard in game theory, if individuals possess heterogeneous benefits from violation (B_i), this universal indifference condition cannot hold. Citizens with a high B_i will have a strict incentive to violate, naturally pushing the system toward pure strategies. However, we do not expect citizens in the real world to literally randomize their decisions in a homogeneous way. Instead, this equilibrium is best interpreted through the lens of Harsanyi's purification theorem. In a population with unobserved heterogeneous preferences, individuals would adopt pure threshold strategies: choosing the good-citizen action S if their private B_i is sufficiently low, and choosing V otherwise. The mixed-strategy compliance probability δ derived in our homogeneous benchmark serves as a tractable mathematical proxy, capturing the aggregate proportion of the population whose private benefits fall below that compliance threshold. Therefore, while individual heterogeneity limits the literal interpretation of mixed strategies, the model's aggregate comparative statics regarding group size remain robust.

7 Conclusions

This study bridges theoretical predictions with experimental findings to examine citizen behaviors in a "Good-Citizen Lottery" framework. Unlike standard voluntary contribution mechanisms where cooperation decays in large groups, our game-theoretic model predicts that the good-citizen lottery leads to asymptotically stable good-citizen behaviors: because the prize pool is funded by violators, the reward for good citizenship scales endogenously with the population size, offsetting the dilution of winning probabilities.

The experimental findings strongly support this prediction. Across both laboratory settings ($n \leq 20$) and a large-scale online extension ($n \in \{50, 100, 200\}$), the proportion of good behavior remained stable as group size increased, confirming the mechanism's scalability. Furthermore, we observe that the *level* of good-citizen behavior consistently exceeds the risk-neutral theoretical benchmark. This "level shift" is driven by a behavioral channel:

individuals who subjectively overestimate small winning probabilities find the uncertain reward relatively more appealing. These insights imply that the citizen lottery is doubly effective: its institutional structure ensures stability against group size, while natural psychological biases (probability weighting) enhance the overall compliance rate.

From a policy perspective, the budget-neutral nature of this mechanism is particularly appealing. It offers a way to minimize public bads without requiring external subsidies or expensive increases in monitoring capacity. Speculatively, if the tendency to overestimate small probabilities is correlated with risk-taking behaviors that exacerbate public bads (e.g., speeding or tax evasion), the citizen lottery could serve as a targeted counterweight. This interplay between behavioral tendencies and institutional design opens promising avenues for future research on the production of public goods and bads.

While the good-citizen lottery is structurally robust in our controlled setting, scaling it into real-world policy introduces substantial institutional frictions. First, success relies heavily on public awareness; the Indian Railways' *Lucky Yatra* pilot ultimately failed due to technical friction and a lack of public understanding, demonstrating that theoretical advantages are moot if citizens do not understand the mechanism. Second, the benchmark model assumes perfect observability. In reality, imperfect monitoring allows 'uncaught violators' to inadvertently dilute the prize pool for genuine compliers. Though this friction likely only slows compliance convergence rather than eliminating it, it demands robust enforcement. Finally, policymakers must navigate the moral framing of gamifying compliance. Unlike our laboratory's neutral gain frame, real-world fines involve sanctioned losses and distinct moral weight. If citizens view the lottery as an inappropriate commodification of civic duty, or distrust the transparency of the prize distribution, the incentive structure will collapse. Therefore, successful field implementation requires pairing this theoretical design with high-visibility communication, equitable monitoring, and careful attention to public acceptability.

References

Andreoni, James, "Impure Altruism and Donations to Public Goods: A Theory of Warm-Glow Giving," *The Economic Journal*, 1990, *100* (401), 464–477.

_ , "Warm-Glow Versus Cold-Prickle: The Effects of Positive and Negative Framing on Cooperation in Experiments," *The Quarterly Journal of Economics*, 1995, *110* (1), 1–21.

— , "Chapter 18 Philanthropy," in Serge-Christophe Kolm and Jean Mercier Ythier, eds., *Applications*, Vol. 2 of *Handbook of the Economics of Giving, Altruism and Reciprocity*, Elsevier, 2006, pp. 1201–1269.

Blau, Benjamin M., R. Jared DeLisle, and Ryan J. Whitby, "Does Probability Weighting Drive Lottery Preferences?," *Journal of Behavioral Finance*, 2020, *21* (3), 233–247.

Charness, Gary and Chun-Lei Yang, "Starting small toward voluntary formation of efficient large groups in public goods provision," *Journal of Economic Behavior & Organization*, 2014, *102*, 119–132.

Duffy, John and Alexander Matros, "On the Use of Fines and Lottery Prizes to Increase Voter Turnout," *Economics Bulletin*, 2014, *34* (2), 966–975.

Fabbri, Marco, Paolo Nicola Barbieri, and Maria Bigoni, "Ride Your Luck! A Field Experiment on Lottery-Based Incentives for Compliance," *Management Science*, 2019, *65* (9), 4336–4348.

Filiz-Ozbay, Emel, Jonathan Guryan, Kyle Hyndman, Melissa S. Kearney, and Erkut Y. Ozbay, "Do lottery payments induce savings behavior? Evidence from the lab," *Journal of Public Economics*, 2015, *126*, 1–24.

Gerardi, Dino, Margaret A. McConnell, Julian Romero, and Leeat Yariv, "Get Out the (Costly) Vote: Institutional Design for Greater Participation," *Economic Inquiry*, 10 2016, *54* (4), 1963–1979.

Gneezy, Uri, *Mixed Signals: How Incentives Really Work*, New Haven: Yale University Press, 2023.

Isaac, R. Mark and James M. Walker, "Group Size Effects in Public Goods Provision: The Voluntary Contributions Mechanism," *The Quarterly Journal of Economics*, 1988, *103* (1), 179–199.

— , — , and Susan H. Thomas, "Divergent Evidence on Free Riding: An Experimental Examination of Possible Explanations," *Public Choice*, 1984, *43* (2), 113–149.

Kearney, Melissa S., Peter Tufano, Jonathan Guryan, and Erik Hurst, "Making Savers Winners: An Overview of Prize-Linked Savings Products," Working Paper 16433, National Bureau of Economic Research October 2010.

Kim, Duk Gyoo, "Vaccination Lottery," *Economics Letters*, 2021, *208*, 110059.

—, "Penalty Lottery," *Scandinavian Journal of Economics*, 2023, *125* (4), 997–1026.

Levitt, Steven D., John A. List, and Sally Sadoff, "The Effect of Performance-Based Incentives on Educational Achievement: Evidence from a Randomized Experiment," NBER Working Papers 22107, National Bureau of Economic Research, Inc Mar 2016.

Morgan, John, "Financing Public Goods by Means of Lotteries," *The Review of Economic Studies*, 2000, *67* (4), 761–784.

— **and Martin Sefton**, "Funding Public Goods with Lotteries: Experimental Evidence," *The Review of Economic Studies*, 2000, *67* (4), 785–810.

Nyqvist, Martina Björkman, Lucia Corno, Damien de Walque, and Jakob Svensson, "Incentivizing Safer Sexual Behavior: Evidence from a Lottery Experiment on HIV Prevention," *American Economic Journal: Applied Economics*, July 2018, *10* (3), 287–314.

Rieger, Marc Oliver, Mei Wang, and Thorsten Hens, "Estimating cumulative prospect theory parameters from an international survey," *Theory and Decision*, 2017, *82*, 567–596.

Snowberg, Erik and Justin Wolfers, "Explaining the Favorite–Long Shot Bias: Is it Risk-Love or Misperceptions?," *Journal of Political Economy*, 2010, *118* (4), 723–746.

Thaler, Richard H. and William T. Ziemba, "Anomalies: Parimutuel Betting Markets: Racetracks and Lotteries," *Journal of Economic Perspectives*, June 1988, *2* (2), 161–174.

Tversky, Amos and Daniel Kahneman, "Advances in prospect theory: Cumulative representation of uncertainty," *Journal of Risk and Uncertainty*, 1992, *5*, 297–323.

Volpp, Kevin G., Leslie K. John, Andrea B. Troxel, Laurie Norton, Jennifer Fassbender, and George Loewenstein, "Financial Incentive–Based Approaches for Weight Loss: A Randomized Trial," *Journal of the American Medical Association*, 12 2008, *300* (22), 2631–2637.

A Appendix: Proofs

Proof of Lemma 1: The most important task is to show $R(\delta,n)=\sum_{k=1}^{n}(1-\delta)^k$. First, we simplify the term $\binom{n}{i}\frac{i}{n+1-i}$.

$$\begin{aligned}\binom{n}{i}\frac{i}{n+1-i} &= \frac{n!}{i!(n-i)!}\frac{i}{n+1-i} = \frac{n+1}{n+1}\frac{n!}{i!(n-i)!}\frac{i}{n+1-i} = \frac{i}{n+1}\frac{(n+1)!}{i!(n+1-i)!}\\ &= \frac{i}{n+1}\binom{n+1}{i}\end{aligned}$$

Substituting this back into the original expression for $R(\delta,n)$:

$$R(\delta,n)=\sum_{i=1}^{n}\frac{i}{n+1}\binom{n+1}{i}\delta^{n-i}(1-\delta)^i$$

To relate this to the expectation of a binomial distribution, add and subtract a term for $i=n+1$.

$$\begin{aligned}R(\delta,n) &= \sum_{i=1}^{n+1}\frac{i}{n+1}\binom{n+1}{i}\delta^{n-i}(1-\delta)^i - \frac{n+1}{n+1}\binom{n+1}{n+1}\delta^{n-(n+1)}(1-\delta)^{n+1}\\ &= \frac{1}{(n+1)\delta}\underbrace{\sum_{i=1}^{n+1} i\binom{n+1}{i}\delta^{n+1-i}(1-\delta)^i}_{=(n+1)(1-\delta)} - \delta^{-1}(1-\delta)^{n+1}\\ &= \frac{1-\delta}{\delta}-\frac{(1-\delta)^{n+1}}{\delta} = \frac{1-\delta}{\delta}(1-(1-\delta)^n),\end{aligned}$$

where the sum $\sum_{i=1}^{n+1} i\binom{n+1}{i}\delta^{n+1-i}(1-\delta)^i$ represents the expected value of a random variable $X\sim\text{Binomial}(n+1,1-\delta)$, which is $(n+1)(1-\delta)$. Finally, let $x=1-\delta$. Since $\delta\in(0,1)$, $x\in(0,1)$. Substituting $\delta=1-x$ yields

$$R(d,n)=\frac{x}{1-x}(1-x^n)=x\frac{1-x^n}{1-x}=x\sum_{k=0}^{n-1}x^k=\sum_{k=1}^{n}x^k,$$

where the geometric series sum formula $\sum_{k=0}^{n-1}x^k=\frac{1-x^n}{1-x}$ was applied. Therefore, $R(\delta,n)$ is simplified as $\sum_{k=1}^{n}(1-\delta)^k$. It is immediate that $R(0,n)=\sum_{k=1}^{n}1^k=n$, and $R(1,n)=\sum_{k=1}^{n}0^k=0$. Since each term in the summation form, $(1-\delta)^k$, is decreasing in δ, $R(\delta,n)$ is a monotone decreasing function in δ. □

Proof of Proposition 1: It is straightforward to show that the solution to $b = \sum_{k=1}^{n}(1-\delta)^k$ is unique for $b \in [0,n]$. Since $\sum_{k=1}^{n}(1-\delta)^k$ is monotone decreasing from n to 0 (Lemma 1) and b is constant in terms of δ, there must be a unique δ^* that equates both hands.

Let $\delta^*(n)$ denote the solution for a given n, such that $b = R(\delta^*(n), n)$. Consider the case for $n+1$. The summation on the right-hand side can be decomposed as:

$$R(\delta, n+1) = \sum_{k=1}^{n+1}(1-\delta)^k = R(\delta, n) + (1-\delta)^{n+1}.$$

Evaluate this function at $\delta = \delta^*(n)$, the solution for n:

$$R(\delta^*(n), n+1) = R(\delta^*(n), n) + (1-\delta^*(n))^{n+1} = b + (1-\delta^*(n))^{n+1}$$

Since $\delta^*(n) \in (0,1)$, the term $(1-d_n^*)^{n+1}$ is strictly positive. Therefore, $R(\delta^*(n), n+1) > b$. By definition, $\delta^*(n+1)$ is the unique value that satisfies $b = R(d^*(n+1), n+1)$. Substituting b, we have

$$R(\delta^*(n), n+1) > R(\delta^*(n+1), n+1).$$

Since $R(\cdot, n+1)$ is a strictly decreasing function in δ, the inequality in function values implies the reverse inequality in the arguments:

$$\delta^*(n) < \delta^*(n+1).$$

Thus, d^* is strictly increasing in n. □

Proof of Proposition 2: In Proposition 1, we established that $\delta^*(n)$ is strictly increasing in n. As $n \to \infty$, the geometric sum converges to $\frac{1-\delta}{\delta}$. Setting $b = \frac{1-\delta}{\delta}$ yields the limit $\lim_{n\to\infty} \delta^*(n) = \frac{1}{1+b}$. Since the sequence is bounded and monotone, it converges. Next, consider the defining equations for population size n and $n+1$:

$$\sum_{k=1}^{n+1}(1-\delta^*(n+1))^k = b = \sum_{k=1}^{n}(1-\delta^*(n))^k.$$

Rearranging the terms:

$$\sum_{k=1}^{n}\left[(1-\delta^*(n+1))^k - (1-\delta^*(n))^k\right] + (1-\delta^*(n+1))^{n+1} = 0.$$

Let $\Delta = \delta^*(n+1) - \delta^*(n)$. Using the Mean Value Theorem on $R(\delta) = \sum_{k=1}^{n}(1-\delta)^k$, there exists

$c \in (\delta^*(n), \delta^*(n+1))$ such that $R(\delta^*(n+1)) - R(\delta^*(n)) = R'(c)\Delta$. Substituting this back gives:

$$R'(c)\Delta + (1-\delta^*(n+1))^{n+1} = 0 \implies \Delta = \frac{(1-\delta^*(n+1))^{n+1}}{-R'(c)}$$

The derivative of $R(\delta)$ with respect to δ at $\delta = c$ is $R'(c) = -\sum_{k=1}^{n} k(1-c)^{k-1}$. Since $c \in (0,1)$, the first term of the sum (where $k = 1$) is -1, and all subsequent terms are negative. Thus, $|R'(c)| > 1$, which provides the inequality:

$$\Delta < (1-\delta^*(n+1))^{n+1}.$$

Since $\delta^*(n)$ is increasing, $\delta^*(n+1) > \delta^*(1)$. For $b \in (0,1)$, the solution for $n = 1$ exists and is given by $(1-\delta^*(1))^1 = b \implies \delta^*(1) = 1-b$. Thus, $1-\delta^*(n+1) < 1-\delta^*(1) = b$. Substituting this into the inequality above yields the proposed bound:

$$\delta^*(n+1) - \delta^*(n) < b^{n+1}$$

This confirms that for $b \in (0,1)$, the sequence stabilizes exponentially fast. □

B Appendix: Experimental Instructions

Welcome

Before you read this instruction, please make sure that you do not force-close this webpage. If you close the webpage, you may not log in to the same experiment again, which may make us unable to calculate your earnings from your participation.

If you have questions or need clarifications, please ask the experimenter via the concurrent Zoom meeting.

Instructions

Thank you for participating in this experiment. Please read carefully the following experiment instructions. At the end of the instructions are quizzes to check your understanding.

Your earnings from this experiment are determined by your choices, other participants' choices, and luck. The monetary unit used in this experiment is called a "token."

Overview

In this experiment, you and many other participants form a group and make simultaneous decisions for 12 times (rounds). In each round, the conditions for your decision would vary, so please be careful in checking the changes. Details follow.

Main task: Choosing a box or a white ball

Participants are randomly assigned to a new group of N people at the beginning of each session.
[Note: The number of group members (N) may vary from round to round, so please make sure to check.]

Participants then simultaneously choose either **a box** or **a white ball**.

- If you choose the box, there is a 70% chance of a blue ball inside the box and a 30% chance of a red ball. You earn 240 tokens with the blue ball, and 40 tokens with the red ball.

- If you choose the white ball, you get 100 tokens. In addition, one randomly selected group member who chose the white ball will receive additional

$$\text{total number of red balls in the group} * 200 \text{ tokens.}$$

Example: Suppose that in a group of four people, two people chose the box and the other two chose the white ball.

- If both of the box choosers get blue balls, no additional earnings go for any of the two white ball choosers.

- If one of the two box choosers gets a red ball, then one of the two white ball choosers receives additional 200 tokens.

- If both box choosers get red balls, one of the two white ball choosers receives additional 400 tokens (=2 red balls*200 tokens).

Feedback at the end of each round

At the end of each round, participants are only told what choices they made in that round. They won't be informed about who the members were or what choices they made during the experiment.

Payments

After the 12 rounds, the server computer will randomly select one round, and the tokens earned by the participant in that round will be converted to cash(*) and paid out.

(*) To be precise, the tokens will be converted into Starbucks e-Gift certificates.

Each round has an equal chance of being selected, so it is of your benefits to play all rounds carefully. The earnings for the selected round will be converted at the rate of 1 token = 100 Korean Won.

Comprehension Check Quiz

You must answer all quizzes correctly to move to the next stage. If necessary, please double-check the instructions above.

Q1 Suppose you belong to a group of 12 people. Which of the following earnings are you *unlikely* to get if you choose the box? (A) 40 tokens (B) 100 tokens (C) 240 tokens

Q2 Suppose you belong to a group of 4 people. Which of the following earnings are you *unlikely* to get if you choose the white ball? (A) 240 tokens (B) 100 tokens (C) 500 tokens

Q3 In a group of 4 people, suppose that members 1 and 2 chose a box and got a blue ball, member 3 chose a box and got a red ball, and member 4 chose a white ball. How much will member 4 earn? [Hint: If only one person chose the white ball, that person will be the one who receives the extra tokens (since one person is randomly selected from a group of one person).] (A) 100 tokens (B) 240 tokens (C) 300 tokens

Q4 Which of the following is a correct description of how the experiment goes? (A) Once the group members are initially determined, the experiment will continue with those members until the end of the experiment. (B) If the box is checked, there is a 60% chance of a red ball. (C) The maximum number of tokens that can be earned by checking the box is 240 tokens.

[After the participant passing the quiz]

You are waiting for all participants to finish reading the instructions and solving the comprehension check quiz. While you are waiting, please review the summary of the experimental procedure below.

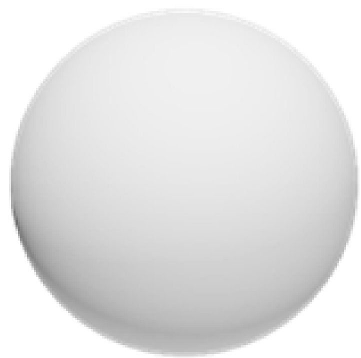
1. The experiment consists of 12 rounds of decision-making.
2. In each round, all participants are randomly assigned to a new group of N people.
3. Each participant simultaneously chooses either a box or a white ball.

- The box contains a blue ball worth 240 tokens with a 70% probability and a red ball worth 40 tokens with a 30% probability.
- If you choose the white ball, you receive 100 tokens. In addition, one randomly selected person among those who chose the white ball receives an additional reward equal to [the number of red balls in the group * 200 tokens].

[Repeat the following for 12 times, with varying N]

In this round, you belong to a group of **6** participants.

Please choose either a box or a white ball. Your choice cannot be undone, so please be careful.



- Choosing the white ball, you receive 100 tokens. In addition, one of those who chose the white ball receives an additional reward equal to [the number of red balls in the group * 200 tokens].
- Choosing the box, you receive 240 tokens if a blue ball is drawn with a 70% chance, or 40 tokens if a red ball is drawn with a 30% chance.

[After the 12 rounds]

The main part of the experiment is complete. Once you fill out the following questionnaire, you will receive the results of your experiment and your earnings.

[After the post-experiment survey]

The results: Round *%R%* was selected as the payout round.

- (When the box was chosen) In this round, you chose the box and got a blue (red) ball. Your reward for this is 240 (40) tokens. The number of red balls that came out of this group is $\%TotalRed\%$. One of the group members who chose the white ball received additional $\%200 * TotalRed\%$ tokens.

- (When a white ball was chosen and the subject was not the winner of the citizen lottery) In this round, you chose the white ball, and your reward for this is 100 tokens. The number of red balls that came out of this group is $\%TotalRed\%$. Unfortunately, you are not one of the members who receive additional $\%200 * TotalRed\%$ tokens.

- (When a white ball was chosen and the subject was the winner of the citizen lottery) In this round, you chose the white ball, and your reward for this is 100 tokens. The number of red balls that came out of this group is $\%TotalRed\%$. You are the one who receives additional $\%200 * TotalRed\%$ tokens.

Your total earning is $\%TotalEarning\%$ tokens or $\%TotalEarning * 100\%$ Korean Won. A Starbucks e-Gift card worth $\%TotalEarning * 100\%$ will be sent to your registered mobile number. You will receive further instructions from Zoom regarding the payment procedure.